

Boolean Algebra

Operations

AND



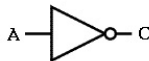
Inputs		Output
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

OR



Inputs		Output
A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

NOT



Input	Output
A	C
0	1
1	0

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Boolean Algebra

What is Boolean Algebra?

Regular algebra is based on numbers (or numerical expressions or variables that represent numbers), and operations that send numbers to numbers.

Boolean algebra is based on the values TRUE and FALSE (or statements or variables that represent the values true or false), and operations that send $\{\text{TRUE}, \text{FALSE}\}$ to $\{\text{TRUE}, \text{FALSE}\}$.

Standard Boolean Operations

Although these operations are based on their usage in English, these tables can be taken as their official definition.

Unary operation:

p	$\neg p$
T	
F	

Binary operations:

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T				
T	F				
F	T				
F	F				

We can then combine these operations. For example, $\neg(p \wedge q)$.

Boolean Algebra

Implies/Converse/Inverse/Contrapositive;
Logical Equivalence

AND



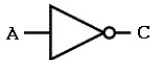
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Implies/Converse/Inverse/Contrapositive

Suppose:

- ▶ If I am making pancakes, the kitchen smells good.

Can I also conclude the following?

- ▶ If the kitchen smells good, then I am making pancakes.
- ▶ If I'm not making pancakes, the kitchen doesn't smell good.
- ▶ If the kitchen doesn't smell good, I am not making pancakes.

Logical equivalence

We can prove our intuition formally using truth tables.

Definition

Two Boolean expressions are **logically equivalent** if they are equal for all possible values of their inputs.

Question: Which expressions above are logically equivalent?

Example proof

Claim

Proof.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
T	T						
T	F						
F	T						
F	F						



Question: Why is this class called *discrete* math?

More practice

Prove the following using a truth table:

► $p \rightarrow q$ is logically equivalent to $\neg p \vee q$

► $p \leftrightarrow q$ is logically equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.

Other truth table proofs

Besides proving logical equivalence, truth tables can be used to prove:

- ▶ two statements are NOT logically equivalent (if their columns are not equal)
 - Ex: $p \rightarrow q \neq q \rightarrow p$
- ▶ a statement is a **tautology** (if it is always TRUE)
 - Ex: $p \vee \neg p$ is a tautology
- ▶ a statement is a **contradiction** (if it is always FALSE)
 - Ex: $p \wedge \neg p$ is a contradiction

Boolean Algebra

Properties

AND



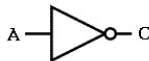
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Boolean Properties

The following logical equivalences can all be proved using truth tables. The properties on this slide you can assume if needed without proof (though you should be able to prove them if asked).

Commutative laws

$$p \wedge q = q \wedge p$$

$$p \vee q = q \vee p$$

Associative laws

$$(p \wedge q) \wedge r = p \wedge (q \wedge r)$$

$$(p \vee q) \vee r = p \vee (q \vee r)$$

Identity laws

$$p \wedge T = p$$

$$p \vee F = p$$

Negation laws

$$p \wedge \neg p = F$$

$$p \vee \neg p = T$$

Boolean Properties cont.

The properties on this slide must be proved first if you want to use them.

DeMorgan's laws

$$\neg(p \wedge q) = \neg p \vee \neg q$$

$$\neg(p \vee q) = \neg p \wedge \neg q$$

Distributive laws

$$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$$

Note: It is also sometimes possible to prove logical equivalence, non-equivalence, tautologies and contradictions using these properties, but we won't cover that method. You are welcome to use it if you want to learn it yourself.

Boolean Algebra

Variations on “Implies”

AND



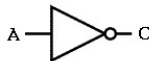
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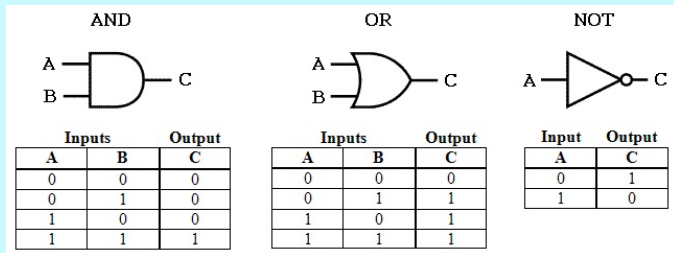
Other ways to say “if-then” in English

If possible, label each phrase below with either $p \rightarrow q$ or $q \rightarrow p$, depending on how the English is typically understood. Try replacing p and q in either order with simple expressions where the implication works only one way (e.g., “ x is a cat”, “ x is an animal”) and see which way sounds correct.

- ▶ p implies q
- ▶ p if q
- ▶ p only if q
- ▶ p is necessary for q
- ▶ p is sufficient for q
- ▶ p because q ($p \therefore q$)
- ▶ p therefore q ($p \therefore q$)
- ▶ p unless q (tricky!)

Further exploration

Here are some standard Boolean operations represented as **logic gates**. :



Logic gates are a core feature of electronics. A single microprocessor may contain more than 100 million logic gates.